

Adeesha Perera - Projects

Machine Learning Engineer | Data Scientist

Production & Portfolio

End-to-end systems built for real-world use

Real-Time Fraud Detection System [LIVE]

Production-grade credit card fraud detection with CatBoost, Feast feature store, Kafka streaming, and automated MLOps lifecycle. Sub-15ms inference latency, ROC-AUC 0.9938, PR-AUC 0.94. 3-tier decision framework: Auto-Block, Manual Review, Soft Audit Signal.

Features: Online/offline feature store with zero training-serving skew, automated drift detection with Evidently AI, weekly retraining pipelines, real-time WebSocket dashboard, Prometheus + Grafana monitoring.

Techno

Hybrid RAG Portfolio Assistant [IN PROGRESS]

Production-grade Retrieval-Augmented Generation system with Docling document extraction, OpenSearch hybrid search (BM25 + k-NN), Cohere reranking, guardrails, and RAGAS evaluation. Deployed on Oracle Cloud via Coolify with webhook-triggered document processing.

Architecture: Parent-child chunking (2000/500 chars), strategy router (direct/rewrite/multi_query/step_back), input/output guardrails, portfolio-only classifier, conversation memory, LRU cache. 6 containers: RAG API, Worker, OpenSearch, MinIO, Docling, OpenSearch Dashboards.

Techno

University Projects

Academic coursework and research projects

Smart Campus Operations Hub

Full-stack campus operations system for facilities, bookings, maintenance tickets, and notifications with JWT + OAuth 2.0 authentication.

Techno

EcoSprout - Carbon Credit Marketplace

Carbon credit marketplace connecting project developers, verifiers, and buyers. Built the educational hub with structured learning modules and community resources.

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Wildlife Safari Project

Full-stack wildlife safari management system with CRUD operations, built as a team project.

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WanderGo

Java web application built as a team project for university coursework.

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Serenity Android App

Native Android application built with Kotlin as a solo university project.

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Traffic Stability Analysis (LAR)

Statistical analysis of Look-Ahead Range impact on traffic flow stability. Hypothesis testing, regression, and visualisation in R.

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ETL & Data Warehousing with Power BI

End-to-end ETL pipeline with Microsoft SSIS, OLAP cube, and Power BI dashboards for business analytics.

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Learning Projects

Practical projects built to learn and experiment

Titanic Survival Prediction

End-to-end ML pipeline with experiment tracking, hyperparameter tuning, and reproducible pipelines.

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Algerian Forest Fire Prediction

Regression models for predicting forest fire severity using linear, ridge, and elastic net regularisation.

Techno

PCA on Breast Cancer Dataset

Principal Component Analysis for dimensionality reduction and visualisation on the Wisconsin breast cancer dataset.

Techno

Logistic Regression Practical

Binary classification with logistic regression on a custom dataset, covering decision boundaries and evaluation metrics.

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Used Car Price Prediction

Regression model comparison: Random Forest, Linear, Ridge, Lasso, and Decision Tree. Random Forest performed best.

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XGBoost Practices

Hands-on practice with XGBoost for both classification and regression tasks.

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Diabetes Prediction

Predicting diabetes progression using Decision Tree Regressor on the sklearn diabetes dataset.

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Fun & Side Projects

Personal experiments and passion projects

Adeesha Perera - Portfolio

Minimalist developer portfolio with AI assistant, dark/light theme, onboarding, and interactive design elements.

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Movie Recommendation System

Comprehensive recommendation engine with KNN, SVD, SBERT content-based, and hybrid ensemble models on MovieLens data.

Techno

F1 Race Predictor

Adeesha Perera - Projects

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Formula 1 race winner prediction using XGBoost regression on historical race data.

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